



Exercises session 5

Class NP

1 Definitions

Definition 1.1 (Verifier) A **verifier** for a language L is a deterministic Turing machine V such that

$$L = \{w \mid V \text{ accepts } \langle w, c \rangle \text{ for some string } c\}$$

A **polynomial time verifier** V runs in time that is polynomial in w . A language is **polynomially verifiable** if it has a polynomial time verifier.

Definition 1.2 (NTIME)

$$NTIME(t(n)) = \{L \mid L \text{ is a language decided by a } O(t(n)) \text{ time non-deterministic TM}\}$$

Definition 1.3 (Class NP) Let's see definitions of the class NP.

1. NP is the class of all languages that have polynomial time verifiers.
2. Defining NP via NTMs

$$NP = \bigcup_k NTIME(n^k)$$

Theorem 1 A language is in NP (has a poly-time verifier) if and only if it is decided by some nondeterministic Turing machine running in polynomial time.

Definition 1.4 (Kleene star operation) Let A be a language. The operation **star** $(*)$ is defined as follows:

$$A^* = \{x_1x_2 \dots x_k \mid k \geq 0 \text{ and each } x_i \in A \text{ where } 1 \leq i \leq k\}$$

2 Session exercises

1. Prove that the class **P** is closed under intersection, union, concatenation and complement.
2. Prove that the class **NP** is closed under union, intersection, kleene star and concatenation.
3. In previous sessions, we have discussed about the following problems

- **Long path**

$\text{LPATH} = \{ \langle G, a, b, k \rangle : \text{there's a simple path from } a \text{ to } b \text{ of length at least } k \}$

- **Subset sum**

$\text{SUBSET-SUM} = \{ \langle A, k \rangle \mid \text{there exists } B \subseteq A \text{ such that } \sum_{b \in B} b = k \}$

Prove that these problems are in NP using both the non-deterministic turing machine and polynomial verifier definition.

4. Given a graph $G = (V, E)$, an **independent set** of G is a set $X \subseteq V$ such that for all $(u, v) \in X \times X$, $(u, v) \notin E$.
 - Give an example of a graph and two non-empty sets of vertices, such that one is an independent set and the other is not
 - Prove that the following problem is in NP

$L = \{ \langle G, k \rangle \mid G \text{ has an independent set of size } k \}$

Solutions

1. Prove that P is closed under intersection, union, concatenation and complement.

• Intersection

Being closed under intersections mean that given $L_1 \in P$ and $L_2 \in P$, then $L_1 \cap L_2 \in P$.

As $L_1 \in P$, then there exist a TM M_1 s.t. $L(M_1) = L_1$; also M_1 runs in polynomial time. As L_2 is also in P , there exist a TM M_2 s.t. $L(M_2) = L_2$ and M_2 also runs in polynomial time.

To decide if a word $w \in \Sigma^*$ is in $L_1 \cap L_2$, we need to check $w \in L_1$ and $w \in L_2$. As we can do this using M_1 and M_2 in polynomial time.

TO-DO: formalize a TM that decides $L_1 \cap L_2$

• Union

We need to prove that

$$L_1 \in P, L_2 \in P \implies L_1 \cup L_2 \in P.$$

As $L_1 \in P$, then there exist TM M_1 s.t. $L(M_1) = L_1$ and M_1 runs in polynomial time. Same with L_2 , there exists a TM M_2 s.t. $L(M_2) = L_2$ and M_2 runs in polynomial time.

In order to decide if $w \in \Sigma^*$ is in $L_1 \cup L_2$, then it should be that either $w \in L_1$, $w \in L_2$, or both. We can decide this in polynomial time using M_1 and M_2 , we can feed w to M_1 , if it accepts, we accept. If not, we feed it into M_2 , if it accepts, we accept, otherwise reject.

TO-DO: formalize this!

• Concatenation

A word $w \in \Sigma^*$ is in $L_1 \cdot L_2$ if $w = w_1 \cdot w_2$ where $w_1 \in L_1$ and $w_2 \in L_2$. We can do this as follows, let's assume $w = w_1 \dots w_m$. Let's also denote $w_0 = w_{m+1} = \epsilon$ (empty word)

For i in $[0, m]$:

check if $w_0 \dots w_i \in L_1$

check if $w_{i+1} \dots w_m \in L_2$

if both, accept.

reject.

TO-DO: prove that this runs in polynomial time!

• Complement.

w belongs to the complement of L if $w \notin L$.

We can decide this using the Turing machine for L , as follows.

if M accepts w :
 reject

else:
 accept.

as M runs in polynomial time, so does above procedure

1. Prove that the class NP is closed under union, and intersection, Kleene star and concatenation.

KLEENE-STAR

- non-deterministically divide w into $x_1 x_2 \dots x_k$
- for each x_i non-deterministically guess the certificate
- if all verifications accept, accept. otherwise, reject.

UNION

- L_1 and L_2 in NP, so there are certificates for both L_1 and L_2 .
Let V_1 and V_2 be such verifiers, then we can build a verifier $V_3 = V_1 \vee V_2$, and this runs in time $\max\{ \text{time}(V_1), \text{time}(V_2) \}$. As both L_1 and L_2 are in NP, this is polynomial.

INTERSECTION

- same as union, but both verifiers need to accept.

CONCATENATION

- similar to the Kleene-star one

• Long path

LPATH = $\{\langle G, a, b, k \rangle : \text{there's a simple path from } a \text{ to } b \text{ of length at least } k\}$

Via NTMs

non-deterministically choose a path of length at least k and at most $|V|$ that goes from a to b . We check in TIME if its a valid simple path

Via Verifier

given a path $\pi_0 \dots \pi_\ell$

- check $\ell \geq k$
- check $\pi_0 = a$ and $\pi_\ell = b$
- check every π_i is unique
- check $(\pi_i, \pi_{i+1}) \in E$.

• Subset sum

SUBSET-SUM = $\{\langle A, k \rangle \mid \text{there exists } B \subseteq A \text{ such that } \sum_{b \in B} b = k\}$

Via NTMs

non-deterministically choose a subset, and verify the sum

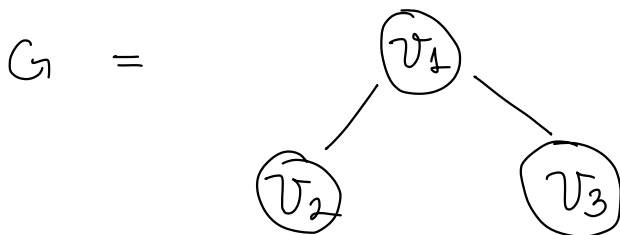
Via Verifier

given a subset $B \subseteq A$, checking if it sums k runs in $\Theta(n)$ with $|A| = n$ and because $|B| \leq |A|$.

3. Given a graph $G = (V, E)$, an **independent set** of G is a set $X \subseteq V$ such that for all $(u, v) \in X \times X$, $(u, v) \notin E$.

- Give an example of a graph and two non-empty sets of vertices, such that one is an independent set and the other is not
- Prove that the following problem is in NP

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Independent sets:

$\{v_1\}$ $\{v_2\}$

$\{v_2, v_3\}$ $\{v_3\}$

non-independent sets

$\{v_1, v_2\}$ $\{v_1, v_3\}$ $\{v_1, v_2, v_3\}$

To prove that is in NP

- We choose non-deterministically a set of vertices of size at least k and at most $|V|$
- We verify if its independent in $O(k^2) \in O(|V|^2)$